

BRAC UNIVERSITY
Department of Computer Science and Engineering

Set: A

Examination: Mid Term
Duration: 1 Hour 20 MinutesSemester: Summer 2025
Full Marks: 25

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. Understanding the question is part of the examination.

Name: _____

1.
CO1A processor executes a program, which has a total of 40×10^6 instructions, categorized as follows:

Instruction Type	Instruction Count	Cycle per instruction
R-type	35% of total instructions	2
I-type	25% of total instructions	3
SB-type	30% of total instructions	4
U-type	10% of total instructions	1

- a) Find the average CPI of the program. 2
- b) If the CPU clock frequency is 2.5 GHz, calculate the execution time of the program. 1

2.
CO1

- a) What is benchmarking, and why is it important in computer performance evaluation? Provide an example of a widely used benchmarking metric. 2
- b) A program runs on CPU X in 150s. CPU Y optimizes 40% of the instructions to run 3 times faster, while the remaining instructions execute at the same speed. Calculate the new execution time of CPU Y. 2

3.
CO1

Read each statement carefully. Determine if the following sentences/statements are true or false. For any false sentence/statement, write its correct form.

- a) 0xA.423 can be stored precisely in the following format.

Sign (1 bit)	Exponent (5 bits)	Fraction (10 bits)
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- b) If a CPU with 1.5x instruction count and 2GHz clock speed performs the same task in 6 seconds as another CPU with 1.0x instruction count and 3GHz clock speed, then the CPI of the first CPU must be lower. 1
- c) A 4-core processor can always achieve a 4x speedup over a single-core version on any program. 1

4.
CO1

$(1.1111011011)_2 \times 2^{23}$. You want to store this floating point number using IEEE754 floating point representation.
You have two format options:

Please Turn Over

	Sign Bit	Exponent	Fraction
Option - 1	1 bit	8 bits	5 bits
Option - 2	1 bit	6 bits	7 bits

- a) If your goal is to achieve precision, which of the two options would you choose? Justify your answer. 2
- b) What is the range of usable actual exponent that can be stored in a 24-bit IEEE format where 16 bits are reserved for the fraction? 2

8.
CO2

- a) Construct the equivalent RISC-V assembly code for the following C program: 5

<pre> arr[20] = func(2, 16); c = arr[21] + d; int func(int a, int b) { if (a <= b) { while (a != b) { p += 2; } b = b & 1; } else { p = b << 2; } return (p + b); } </pre>	Variables c and d are stored in registers X₁₈ and X₁₉ respectively. Function arguments a and b are stored in registers X₁₀ and X₁₁ respectively. Variable p in the function func, also uses register X₁₀. arr is a halfword array. The base address of arr is stored in X₃₀. Hint: << means left shift
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b)

Instruction Address	Instruction
0xB0A1834A	beq X ₂₇ , X ₂₂ , L0
.....	
.....	
.....	
0xB0A1838A	L0: add X ₂ , X ₆ , X ₇

Convert the instruction, beq X₂₇, X₂₂, L0; to machine code. For beq, opcode = 1100111, funct3 = 000. Your final answer must be in Hexadecimal.

- c) Construct the equivalent RISC-V assembly code for the following C program: 3

<pre> a = 0xD835B12C; b = a / 64; </pre>	a and b variables are stored in X₁₆, X₂₀ registers, respectively.
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